

Programming Fundamental

Experiment # 12

Switch Statement

1. Objective/s:

- Switch Statement
- Rules to Implement Switch Statement
- Examples
- Lab Task

2. Theory:

A switch statement allows a variable to be tested for equality against a list of values. Each value is called a case, and the variable being switched on is checked for each case.

2.1. Syntax:

The syntax for a switch statement in C++ is as follows –

```
switch(expression) {  
    case constant-expression :  
        statement(s);  
    break; //optional  
    case constant-expression :  
        statement(s);  
        break; //optional  
  
    // you can have any number of case statements.  
    default : //Optional  
        statement(s);  
}
```

2.2. Example:

```
#include <iostream>  
using namespace std;
```

```
int main(){
    int num=5;
    switch(num+2) {
        case 1:
            cout<<"Case1: Value is: "<<num<<endl;
        case 2:
            cout<<"Case2: Value is: "<<num<<endl;
        case 3:
            cout<<"Case3: Value is: "<<num<<endl;
        default:
            cout<<"Default: Value is: "<<num<<endl;
    }
    return 0;
}
```

3. Rules:

The following rules apply to a switch statement

- The **expression** used in a **switch** statement must have an integral or enumerated type, or be of a class type in which the class has a single conversion function to an integral or enumerated type.
- You can have any number of case statements within a switch. Each case is followed by the value to be compared to and a colon.
- The **constant-expression** for a case must be the same data type as the variable in the switch, and it must be a constant or a literal.
- When the variable being switched on is equal to a case, the statements following that case will execute until a **break** statement is reached.
- When a break statement is reached, the switch terminates, and the flow of control jumps to the next line following the switch statement.
- Not every case needs to contain a break. If no break appears, the flow of control will *fall through* to subsequent cases until a break is reached.
- A **switch** statement can have an optional **default** case, which must appear at the end of the switch. The default case can be used for performing a task when none of the cases is true. No break is needed in the default case.

4. Lab Tasks

Write a program using switch statement that takes numbers from user and prints grade. Grade division is as follows. only listed marks can be assigned

90 A

80 B

70 C

60 D

50 E

< 50 F

Design a simple calculator using switch that prompts the user to select operator and performs the selected operation. Output should look like:

Enter two numbers

.....

Enter '+' for addition

Enter '-' for subtraction

Enter '*' for multiplication

Enter '/' for division

Enter your choice

.....